

Basic coding problems -

1. Reverse a String
2. Check for Palindrome
3. Find the Maximum Element in an Array
4. Count the Number of Vowels in a String
5. Sum of Digits of a Number
6. Fibonacci Number (nth term)
7. Print All Prime Numbers up to N
8. Remove Duplicates from a String
9. Second Largest Element in an Array
10. Check if Two Strings Are Anagrams
11. Write a program for leap year.

Topic Wise DSA problems -

Array & String-

1. Contains duplicate - <https://leetcode.com/problems/contains-duplicate/description/>
2. Kadane
algorithms-<https://www.geeksforgeeks.org/problems/kadanes-algorithm-1587115620/1>
3. Best time to buy sell stock
-<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/>
4. Two sum pair
-<https://www.geeksforgeeks.org/problems/count-pairs-with-given-sum5022/1>
5. Subarray with zero
sum-<https://www.geeksforgeeks.org/problems/subarray-with-0-sum-1587115621/1>
6. Two sum-<https://leetcode.com/problems/two-sum/description/>
7. Triplet sum- <https://www.geeksforgeeks.org/problems/triplet-sum-in-array-1587115621/1>
8. Best time to buy and sell stock-
<https://leetcode.com/problems/best-time-to-buy-and-sell-stock/description/>
9. Move all negative numbers to one side -
10. <https://www.geeksforgeeks.org/problems/move-all-negative-elements-to-end1813/1>
11. Valid anagram-<https://leetcode.com/problems/valid-anagram/description/>
12. Reverse words -
<https://www.geeksforgeeks.org/problems/reverse-words-in-a-given-string5459/1>
13. Min Number of Flips-<https://www.geeksforgeeks.org/problems/min-number-of-flips3210/1>
14. Longest common prefix-
<https://leetcode.com/problems/longest-common-prefix/description/>
15. Smallest Distinct Window-
<https://www.geeksforgeeks.org/problems/smallest-distant-window3132/1>
16. Print anagram together-
<https://www.geeksforgeeks.org/problems/print-anagrams-together/1>

- 17. Replace element with greatest element on right side-<https://leetcode.com/problems/replace-elements-with-greatest-element-on-right-side/description/>
- 18. Missing and repeated value-<https://leetcode.com/problems/find-missing-and-repeated-values/description/>
- 19. Reverse string-<https://leetcode.com/problems/reverse-string/description/>
- 20. Valid palindrome-<https://leetcode.com/problems/valid-palindrome/description/>
- 21. Sliding window-<https://leetcode.com/problems/number-of-sub-arrays-of-size-k-and-average-greater-than-or-equal-to-threshold/description/>
- 22. Pattern programs-
<https://www.geeksforgeeks.org/pattern-programs-in-c/>

Stack & Queue-

- 23. Valid parentheses-<https://leetcode.com/problems/valid-parentheses/description/>
- 24. Implement stack using queue-<https://leetcode.com/problems/implement-stack-using-queues/description/>
- 25. Implement queue using stack-<https://leetcode.com/problems/implement-queue-using-stacks/description/>
- 26. Next greater element-<https://www.geeksforgeeks.org/problems/next-larger-element-1587115620/1>

Binary Search-

- 27. Binary search-<https://leetcode.com/problems/binary-search/description/>
- 28. Search in 2d matrix-<https://leetcode.com/problems/search-a-2d-matrix/description/>
- 29. First and last occurrence-<https://www.geeksforgeeks.org/problems/first-and-last-occurrences-of-x3116/1>

Priority Queue-

- 30. Kth largest element(Priority queue)-<https://leetcode.com/problems/kth-largest-element-in-an-array/description/>

Linked list-

- 31. Linked list operations-<https://www.programiz.com/dsa/linked-list-operations>
- 32. Reverse linked list-<https://leetcode.com/problems/reverse-linked-list/description/>
- 33. Linked list cycle-<https://leetcode.com/problems/linked-list-cycle/description/>
- 34. Palindrome linked list-<https://leetcode.com/problems/palindrome-linked-list/description/>
- 35. Remove duplicates from sorted linked list-<https://leetcode.com/problems/remove-duplicates-from-sorted-list/description/>
- 36. Middle of linked list-

- 37. <https://leetcode.com/problems/middle-of-the-linked-list/description/>
- 38. Segregate even and odd nodes in linked list-<https://www.geeksforgeeks.org/problems/segregate-even-and-odd-nodes-in-a-linked-list5035/1>
- 39. Delete node having greater value on right-<https://www.geeksforgeeks.org/problems/delete-nodes-having-greater-value-on-right/1>

Matrix-

- 40. Transpose Matrix - <https://leetcode.com/problems/transpose-matrix/description/>
- 41. Spiral traversal of matrix-<https://www.geeksforgeeks.org/problems/spirally-traversing-a-matrix-1587115621/1>
- 42. Rotate matrix by 90 degrees-<https://www.geeksforgeeks.org/rotate-a-matrix-by-90-degree-in-clockwise-direction-without-using-any-extra-space/>

Trees-

- 43. Binary Tree inorder,preorder,postorder traversal method.
- 44. Depth of binary tree-<https://leetcode.com/problems/maximum-depth-of-binary-tree/description/>
- 45. Leaf at same level-<https://www.geeksforgeeks.org/problems/leaf-at-same-level/1>
- 46. Balance tree check-<https://www.geeksforgeeks.org/problems/check-for-balanced-tree/1>
- 47. Sum tree-<https://www.geeksforgeeks.org/problems/sum-tree/1>
- 48. Sum of nodes on the longest path-<https://www.geeksforgeeks.org/problems/sum-of-the-longest-bloodline-of-a-tree/1>
- 49. Kth ancestor of node in binary tree-
<https://www.geeksforgeeks.org/kth-ancestor-node-binary-tree-set-2/>
- 50. Find a value in BST-<https://www.geeksforgeeks.org/problems/search-a-node-in-bst/1>
- 51. Check for BST-<https://www.geeksforgeeks.org/problems/check-for-bst/1>
- 52. Binary tree to BST-<https://www.geeksforgeeks.org/problems/binary-tree-to-bst/1>
- 53. Brothers from different roots -
<https://www.geeksforgeeks.org/problems/brothers-from-different-root/1>

Graph-

- 54. BFS of graph - <https://www.geeksforgeeks.org/problems/bfs-traversal-of-graph/1>
DFS of graph - <https://www.geeksforgeeks.org/problems/depth-first-traversal-for-a-graph/1>
- 55. Detect cycle in undirected graph -
<https://www.geeksforgeeks.org/problems/detect-cycle-in-an-undirected-graph/1>
- 56. Flood fill - <https://leetcode.com/problems/flood-fill/description/>
- 57. Number of operations to make network connected-<https://leetcode.com/problems/number-of-operations-to-make-network-connected/description/>

58. Number of islands-

<https://www.geeksforgeeks.org/problems/find-the-number-of-islands/1>

59. Topological sort-<https://www.geeksforgeeks.org/problems/topological-sort/1>

Dynamic Programming-

60. Coin change -<https://www.geeksforgeeks.org/problems/coin-change2448/1>

61. Partion equal subset sum-

<https://www.geeksforgeeks.org/problems/subset-sum-problem2014/1>

62. 0-1 knapsack-<https://www.geeksforgeeks.org/problems/0-1-knapsack-problem0945/1>

63. Longest common

subsequence-<https://www.geeksforgeeks.org/problems/longest-common-subsequence-1587115620/1>

64. Longest palindromic

substring-<https://leetcode.com/problems/longest-palindromic-substring/description/>

65. Edit distance-<https://www.geeksforgeeks.org/problems/edit-distance3702/1>

66. <https://www.geeksforgeeks.org/count-subsequences-product-less-k/>

67. Gold mine problem-<https://www.geeksforgeeks.org/problems/gold-mine-problem2608/1>

Bit Manipulation-

68. Number of 1 bits-<https://www.geeksforgeeks.org/problems/set-bits0143/1>

69. Missing number($O(1)$ space)-<https://leetcode.com/problems/missing-number/description/>

70. Power of 2 (without using modulo

%) -<https://leetcode.com/problems/power-of-two/description/>

71. Minimum bit flip to convert number -

<https://leetcode.com/problems/minimum-bit-flips-to-convert-number/description/>

72. How will you check number is odd or even without using modulo operator.

73. Position of set bit-<https://www.geeksforgeeks.org/problems/find-position-of-set-bit3706/1>

74. Toggle bits -<https://www.geeksforgeeks.org/problems/toggle-bits-given-range0952/1>

Greedy-

75. Job sequencing

problem-<https://www.geeksforgeeks.org/problems/job-sequencing-problem-1587115620/1>

76. Huffman coding-<https://www.geeksforgeeks.org/problems/huffman-encoding3345/1>

77. Fractional

Knapsack-<https://www.geeksforgeeks.org/problems/fractional-knapsack-1587115620/1>

78. Minimum

Platforms-<https://www.geeksforgeeks.org/problems/minimum-platforms-1587115620/1>

Data Structures-

What is linear and non linear data structure?

What is a stack data structure? What are the operations available on stack?

What is a Queue data structure? What are the operations available on stack?

What is array data structure?

What is a linked list and its types?

What are the advantages of a linked list over an array? In which scenarios do we use Linked List and when Array?

What is an asymptotic analysis of an algorithm?

What is a hashmap in data structure? and its time complexity.

What is binary tree data structure?

What is binary search tree data structure?

What are tree traversals? Inorder, preorder, postorder.

What is a priority queue?

What is graph data structure and its representations?

What is the difference between the Breadth First Search (BFS) and Depth First Search (DFS)?

What is AVL tree data structure

Define Red-Black Tree

What is a heap data structure?

Which sorting algorithm is considered the fastest? Why?

Learn best case, average case, worst case time complexity of Bubble sort, selection sort, insertion sort, merge sort, quick sort.

What is the merge sort? How does it work?

What are the advantages of a binary search over a linear search?

What is the time complexity of binary search?

What is hashing? What are the different techniques of hashing.

What is the time complexity of search, insert, and delete operations in a Hash Table?

What is recursion?

What is the time complexity to search in an unsorted and sorted array?

What is time complexity of Linked List operations? (insertion, deletion, search, traversal)

What approach would you take to find the shortest route between two locations?(Dijkstra algorithm)

What is the minimum spanning tree?

Prim's algorithm, kruskal algorithm?

DBMS-

What is the difference between **WHERE** and **HAVING**?

Write a query to fetch the second highest salary from a table.(Using subquery/Limit offset/Rank)

Inner Join/Left Join/Right Join/Outer Join

Find the number of employees in each department.

What are keys (Primary key, Candidate Key, Alternate Key, Foreign Key, Composite key, Super key,)

What are **PRIMARY KEY** and **UNIQUE** constraints?

What is a **VIEW** in SQL?

Write a query to fetch top 3 highest-paid employees.(Use limit offset/Rank)

Difference between **DELETE**, **TRUNCATE**, and **DROP**?

What is an entity-relationship diagram (ERD)?

What is normalization? Why is it important?What are the various forms of Normalization?

Explain different types of relationships amongst tables in a DBMS.

What are tuples and attributes in DBMS?

Find employees who earn more than the average salary.

Difference between **UNION** and **UNION ALL**?

What is the **DISTINCT** keyword in sql?

What is the purpose of the **GROUP BY** clause in SQL?

What are indexes?What is the difference between a clustered and non-clustered index?

What are ACID properties in a transaction?

What is Pattern Matching in SQL?(Wildcard)

What are the differences between OLTP and OLAP?

- DDL (Data Definition Language): CREATE, ALTER, DROP, TRUNCATE.
- DML (Data Manipulation Language): SELECT, INSERT, UPDATE, DELETE.

- DCL (Data Control Language): GRANT, REVOKE.
- TCL (Transaction Control Language): COMMIT, ROLLBACK, SAVEPOINT.

Operating System-

What is a bootstrap program in OS?

What is different between main memory and secondary memory

What is thrashing in OS?

What is the difference between multitasking and multiprocessing OS?

What is the difference between paging and segmentation?

Write a difference between internal fragmentation and external fragmentation?

What is virtual memory?

What is thread in OS?

What is a process? What are the different states of a process?

What is the difference between process and thread?

What are various sections of the process?

What is a Scheduling Algorithm? Explain different types of scheduling algorithms. Which algorithm gives best throughput?

What do you mean by process synchronization? Explain process synchronization techniques.

What is the critical section?

What is preemptive and non preemptive scheduling?

What is Kernel in OS?

What do you mean by Semaphore in OS? What are its types?

Explain the difference between semaphore and mutex?

What is Context Switching?

What is a deadlock in OS? What are the necessary conditions for a deadlock?

What is starvation and how it is different from deadlock?

What do you mean by Belady's Anomaly?

Explain demand paging?

OOPS Concepts-

What are the main features of OOPs?

What are access specifiers and what is their significance?

What are the various types of inheritance? Explain.

What is meant by polymorphism? Write its types.

What is the difference between overloading and overriding?
What is data abstraction ?
What is an abstract class?
What is a class?What is an object?
What is encapsulation?
What is a constructor?
What are the various types of constructors?
What is a copy constructor?
What is a destructor?
What is the difference between a structure and a class in C++?
What are friend functions
What is the virtual function and pure virtual function?
C++ Static Data Members
What is enumeration?
What is this pointer?

Computer Network-

Define different types of network topology
What is DNS?
What is the use of a router and how is it different from a gateway?
What is the SMTP protocol?
Define the 7 different layers of the OSI Reference Model. Explain protocol associated with each layer
What are the HTTP and the HTTPS protocol?
What is TCP and UDP protocol?Compare.
What is the ICMP protocol?
What is the ARP protocol?
What is the MAC address and IP address.Compare.
What is the port number? What is socket?
What is a subnetting?
Compare the hub vs switch
What is the firewall?

What are the different classes of IPv4?

What are unicasting, multicasting, broadcasting, and anycasting?

Identify the Class, Network IP Address, Direct broadcast address and Limited broadcast address of each IP Address.

- | | |
|----|------------|
| 1. | 1.2.3.4 |
| 2. | 10.15.20.6 |

Which protocols work with application layers?

How many hosts will be in a /29 subnet?

Language specific problems-

You should prepare for (C,C++,python) or (C,Java,python)

C - <https://www.interviewbit.com/c-interview-questions/>

C++ - <https://www.interviewbit.com/cpp-interview-questions/>

Python - <https://www.interviewbit.com/python-interview-questions/>

Java - <https://www.interviewbit.com/java-interview-questions/>